

2 Three supply regimes isolate arrivals before search

Takeaway. Pages 1–2 solve the closed, known-one-driver benchmark. With latent Poisson supply, we first ask what a new period-2 cohort changes at the same fixed footprint; only then do we let the platform expand notification reach.

Regime	Commitment	Drivers who can volunteer after first-window failure	Added object
I. Incumbents only	(p_1, p_2)	Retained first-window rejectors	Recall value
A. Fixed footprint + arrivals	(p_1, p_2)	Retained rejectors plus an independent fresh core cohort	Replenishment value
E. Expanded search	(p_1, p_2, s)	Retained rejectors plus the same fresh core cohort, and an outer-annulus cohort in rescue	Incremental reach value

2.1 Common Poisson environment

The two response windows have equal length and the core footprint has area one. The first-window incumbents and second-window core arrivals are

$$N_1 \sim \text{Pois}(m), \quad N_2^C \sim \text{Pois}(m), \quad N_1 \perp N_2^C. \quad (5)$$

Thus m is a *per-window cohort flow*, not a stock counted twice. Every incumbent in N_1 sees the focal order and the full commitment before choosing accept or reject-and-wait. Fresh drivers in N_2^C first see it only if a terminal notification is sent. A rejector remains physically available for the terminal lottery with exogenous probability $\omega \in [0, 1]$.

Throughout this numerical framework, rider value v and persistent driver cost c satisfy $v, c \sim U[0, 1]$; write F for the driver-cost CDF. Rider and incumbent delay factors are $\beta, \delta \in (0, 1]$; $\tau \geq 0$ is the outer pickup-cost slope and $\bar{s} \geq 1$ is the search cap.

2.2 What is, and is not, nested

Regimes I and A are two supply benchmarks: one has no new cohort; the other has time-homogeneous replenishment. They should not be called nested mechanisms unless the platform can suppress the core broadcast. Their policy domains coincide, while expanded search reproduces the fixed-arrival technology exactly at $s = 1$:

$$\begin{aligned} \mathcal{P}_I = \mathcal{P}_A &= \{(p_1, p_2) : 0 \leq p_1 \leq p_2 \leq 1\}, \\ \mathcal{P}_E &= \{(p_1, p_2, s) : 0 \leq p_1 \leq p_2 \leq 1, 1 \leq s \leq \bar{s}\}. \end{aligned} \quad (6)$$

Within every regime, $p_2 = p_1$ remains the flat-price restriction. All comparisons below re-optimize the available policy; no common fixed p_1 is imposed. On an active continuation branch the computation restricts $p_2 \leq \beta$ without loss, since no rider can obtain positive delayed surplus above β ; the inactive branch $p_1 \geq \beta$ is checked separately.

3 A complete cutoff-WPBE sits inside every design problem

Takeaway. The platform commits first. Every candidate policy μ_r induces driver strategies, rider continuation, and Bayesian beliefs; the platform compares policies only after solving that full cutoff-WPBE correspondence.

Timing and information	Terminal supply objects
0 Holding $r \in \{I, A, E\}$ fixed, the platform commits to $\mu_I = \mu_A = (p_1, p_2)$ or $\mu_E = (p_1, p_2, s)$.	Let $a \in [0, p_1]$ be the first-window cutoff (accept iff $c \leq a$), write $[x]_+ = \max\{x, 0\}$, and let $j = 1$ denote repeat, $j = 2$ rescue. Conditional on universal rejection, retained incumbent volunteer intensity is
1 A rider posts iff $v \geq p_1$. All N_1 incumbents see the order, prices, and own c , but not the count or other costs.	$I_j^\omega(a) = \omega m[F(p_j) - F(a)]_+$. Fresh volunteer intensities in regime r are
2 After universal rejection, each rejector remains available with probability ω . The rider observes failure, not realized driver counts, and chooses abandon, repeat p_1 , or rescue p_2 .	$e_j^I = 0, \quad e_j^A = mF(p_j),$ $e_1^E = mF(p_1), \quad e_2^E = mF(p_2) + m \int_1^s F(p_2 - d(u)) du.$
3 The chosen notification is sent. Eligible rejectors and reached fresh volunteers enter one anonymous lottery.	Here $d(u) \geq 0$ is the post-assignment outer pickup cost defined on page 6. For every regime, $\Lambda_j^r(a) = I_j^\omega(a) + e_j^r, \quad C_j^r(a) = 1 - e^{-\Lambda_j^r(a)}. \quad (7)$ Λ_j^r is the expected total willing-driver count, not a per- m rate. The rider compares 0, $C_1^r(\beta v - p_1)$, and $C_2^r(\beta v - p_2)$. Let $\eta_j^r(a)$ be the resulting action probability conditional on posting and first-window failure.

3.1 Pure cutoff-WPBE test

Let $h(z) = (1 - e^{-z})/z$ for $z > 0$ and $h(0) = 1$. A type- c incumbent's accept-minus-wait payoff under a proposed aggregate cutoff is

$$D_r(c; a) = h(mF(a))(p_1 - c) - \delta \omega e^{-mF(a)} \sum_{j=1}^2 \eta_j^r(a) h(\Lambda_j^r(a)) [p_j - c]_+. \quad (8)$$

Together with the rider and fresh-driver strategies above, a is a cutoff-WPBE iff on-path beliefs obey Bayes' rule and

$$D_r(c; a) \geq 0 \quad \forall c < a, \quad D_r(c; a) \leq 0 \quad \forall c > a, \quad (9)$$

with the marginal type indifferent at an interior cutoff. Because D_r is piecewise affine in c , the solver checks the entire type domain at its affine interval endpoints, not on a sampled type grid.

3.2 Outer mechanism design

Ex-ante completion and expected outer contacts are

$$\begin{aligned} M_r(\mu_r, a) &= (1 - p_1) \left[1 - e^{-mF(a)} + e^{-mF(a)} \{ \eta_1^r(a) C_1^r(a) + \eta_2^r(a) C_2^r(a) \} \right], \\ Q_E^O(\mu_E, a) &= (1 - p_1) e^{-mF(a)} \eta_2^E(a) m(s - 1), \quad Q_I^O = Q_A^O = 0. \end{aligned} \quad (10)$$

Let $B > 0$ be value per completion, $\kappa \geq 0$ the incremental outer-contact cost, $\theta = (m, \beta, \delta; \tau, \omega, \bar{s}, B, \kappa)$, and $\mathcal{E}_r^{\text{WPBE}}(\mu_r; \theta)$ the set of cutoffs satisfying (9). For $J_r = BM_r - \kappa Q_r^O$, the target design problem is

$$V_r(\theta) = \max_{\mu_r \in \mathcal{P}_r} \min_{a \in \mathcal{E}_r^{\text{WPBE}}(\mu_r; \theta)} J_r(\mu_r, a), \quad \text{fix } p_1 \longrightarrow \max_{p_2, s} \longrightarrow \max_{p_1}. \quad (11)$$

Conservative equilibrium selection minimizes the same J_r used outside. Reported candidates combine this nested profile with an independent global-basin search and dense WPBE recertification.

4 Fixed footprint: what do fresh arrivals change?

Takeaway. Regimes I and A give the platform the same price menu (p_1, p_2) . Their only difference is whether a new, independent core cohort can first see the order after first-window failure.

Same cross-checked (p_1, p_2) design; only period-2 supply changes

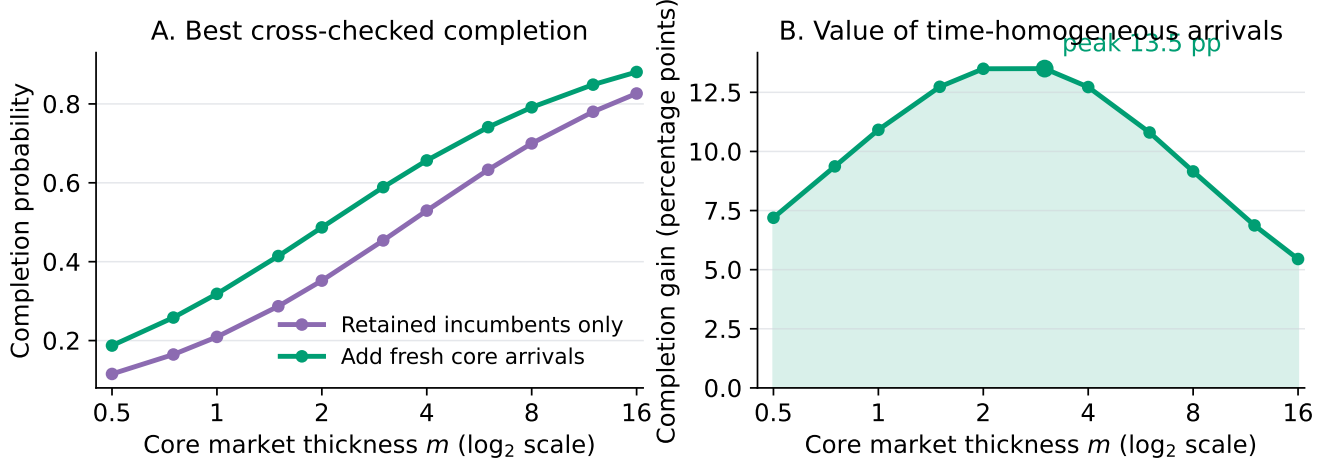


Figure 3: Incumbents only versus fixed footprint with time-homogeneous fresh arrivals. Every point is the best cross-checked candidate from the nested profile and an independent global-basin search, with its complete cutoff-WPBE re-solved. Main slice: $(\beta, \delta, \omega) = (.8, .8, .8)$.

The supply decomposition is deliberately minimal:

$$\Lambda_j^I(a) = \omega m [F(p_j) - F(a)]_+, \quad \Lambda_j^A(a) = \Lambda_j^I(a) + m F(p_j). \quad (12)$$

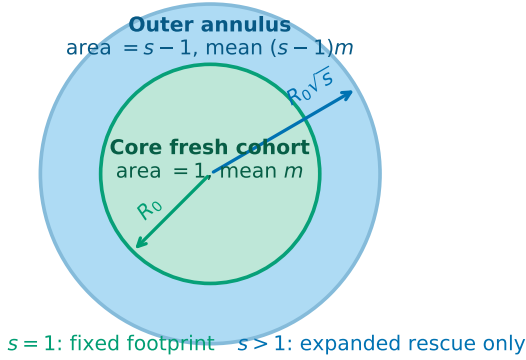
The first term is the cost-screened set of rejectors who remain in the market; the second is the new core cohort after price-based thinning. Setting $\omega = .8$ means the plotted benchmark has genuine incumbent exit: twenty percent of rejectors become unavailable before the terminal action.

Fresh arrivals have three equilibrium effects. They directly increase terminal coverage; their lottery competition lowers an incumbent's chance of winning after waiting; and the resulting coverage changes which riders choose repeat or rescue. Therefore the green curve is not obtained by adding $mF(p_j)$ to an old fixed-price outcome. Both prices and the first-window cutoff are re-solved at every m .

Identification. $V_A - V_I$ is the completion value of time-homogeneous replenishment at a fixed geographic footprint. It is not yet the value of searching farther away.

5 Expanded search adds only an outer annulus

Takeaway. The core cohort remains exactly the one in regime A. Expanded rescue changes only who outside that core can first learn about the failed order; s is an area multiplier, not an arrival probability.



Normalize the core radius to R_0 . A committed multiplier $s \geq 1$ means

$$R(s) = R_0\sqrt{s}, \quad N_2^O(s) \sim \text{Pois}((s-1)m), \quad (13)$$

where $N_2^O(s)$ is independent of the core cohort N_2^C . Equivalently, the combined fresh pool has mean sm , and its core component is obtained by ordinary Poisson thinning with share $1/s$.

Repeat uses core reach $s_1 = 1$. Only the announced rescue branch uses $s_2 = s$.

5.1 Driver-paid pickup friction occurs after assignment

Let ℓ be driver-order pickup distance and $u = (\ell/R_0)^2 \in [1, s]$ the outer driver's area rank. Core pickup is absorbed into persistent cost c ; an outer winner additionally pays

$$d(u) = \tau(\sqrt{u} - 1)_+. \quad (14)$$

The fresh driver knows u, p_2, c before volunteering but pays $d(u)$ only when selected. Her rescue response payoff is

$$h(\Lambda_2^E(a))[p_2 - c - d(u)].$$

Because the same positive assignment probability multiplies revenue and cost,

$$\text{volunteer} \iff c + d(u) \leq p_2. \quad (15)$$

Spatial independent increments plus mark thinning therefore give rescue outer-volunteer intensity

$$e_O(p_2, s) = m \int_1^s F(p_2 - d(u)) du, \quad \frac{\partial e_O}{\partial s} = mF(p_2 - d(s)). \quad (16)$$

For $F = U[0, 1]$, increasingly distant rings contribute fewer volunteers and eventually none. There is no notification-time relocation cost and hence no fresh-entry fixed point in this maintained benchmark.

Separation of roles. p_2 changes willingness among all reached drivers. s changes who can see rescue. Driver pickup cost $d(u)$ thins willing supply, but it is not a platform cost of sending notifications.

6 A platform search cost makes $s = 1$ an economic choice

Takeaway. A completion-only objective with free contacts asks for a maximal coverage frontier and often pushes s to its cap. To ask whether expansion is worth doing, the objective must price the incremental outer contacts.

Object	Who bears it?	Economic role
$d(u)$	Assigned outer driver	Enters the response threshold and thins willing supply.
κ per outer contact	Platform/system	Prices notification attention and incremental dispatch capacity.

The expanded-search rescue branch is executed ex ante with probability

$$q_R^E = (1 - p_1)e^{-mF(a)}\eta_2^E(a).$$

Thus expected extra contacts and the maintained design objective are

$$Q_E^O = q_R^E m(s - 1), \quad J_E^\kappa = BM_E - \kappa Q_E^O. \quad (17)$$

This is completion net of notification opportunity cost, not platform profit; calling it profit would require separate rider fares and driver wages.

Incremental outer-contact cost turns reach into an economic choice

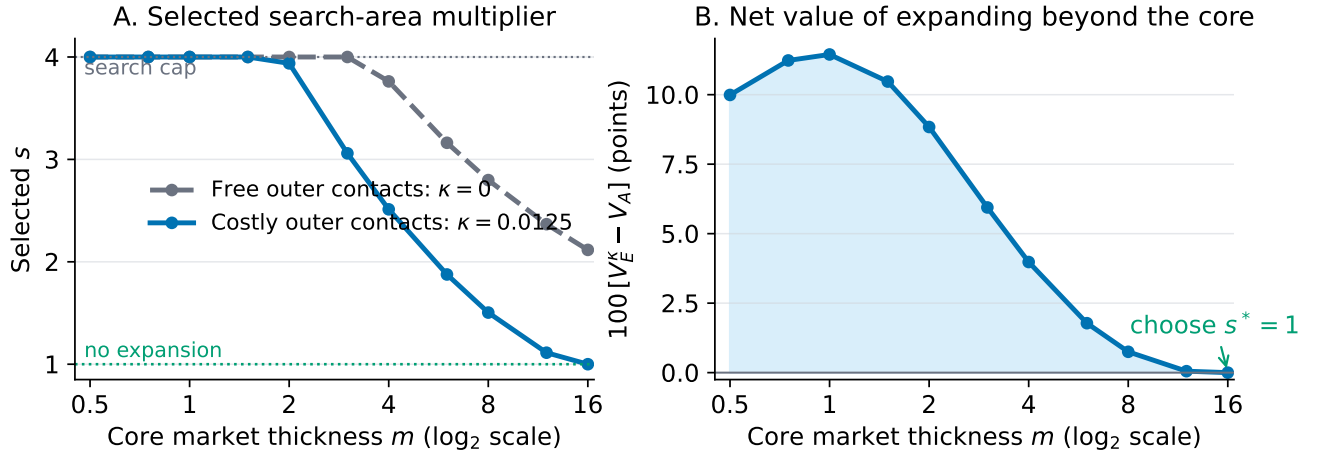


Figure 4: Free-search frontier versus the costly-search design. Both lines cross-search (p_1, p_2, s) and re-solve the inner cutoff-WPBE at each m . The main slice uses $(\beta, \delta) = (.8, .8)$, $B = 1$, $\kappa = .0125$, $\tau = .25$, $\omega = .8$, and $\bar{s} = 4$; $s = 1$ exactly reproduces fixed-footprint arrivals.

The decline in $s^*(m)$ is not imposed. As the optimized core becomes more reliable, the marginal completion value of another ring falls while its expected contact cost remains. The outer design can therefore select a cap, an interior multiplier, or no expansion $s^* = 1$.

7 Market conditions and the research boundary

Takeaway. Fresh arrivals matter most in an intermediate core market. Costly geographic expansion is most valuable when the core is thin and can optimally switch off once fixed-footprint rescue is reliable.

Cross-search (p_1, p_2, s) and re-solve its cutoff-WPBE in every cell

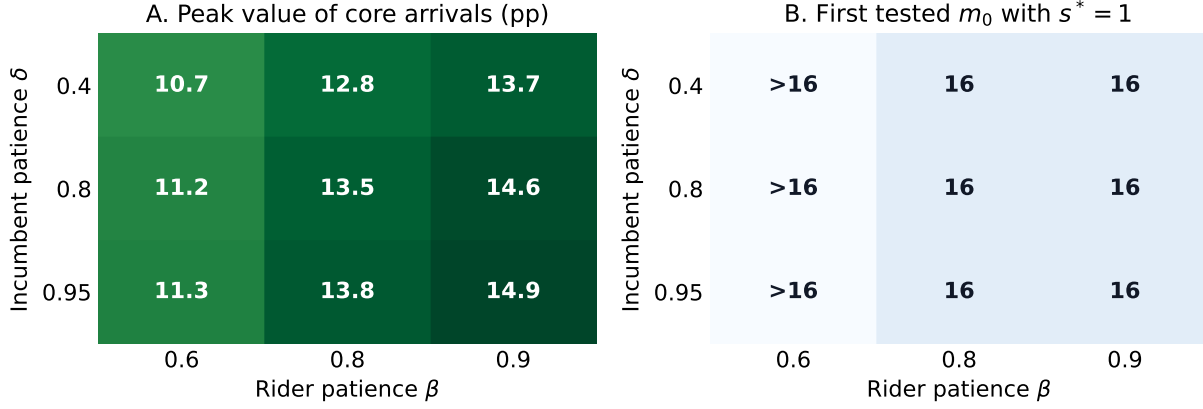


Figure 5: Comparative statics across rider patience β and incumbent patience δ . Each cell uses the same m grid and separately re-solves all three outer mechanism-design problems and their complete cutoff-WPBE correspondences. Other parameters match Figure 4. Here > 16 means search does not switch off anywhere on the tested m grid.

Layer	What it identifies	Maintained cost normalization
I $\rightarrow$ A	Value of an independent, time-homogeneous core cohort at fixed footprint	Core broadcast is installed; report the best cross-checked completion.
A $\rightarrow$ E	Incremental value of outer geographic reach	Charge κ per actually executed outer contact.

If activation of the core cohort is itself a platform choice, let $\kappa_C, \kappa_O \geq 0$ be the core- and outer-contact costs. Suppressing regime superscripts, the symmetric extension is

$$J_{\kappa_C, \kappa_O} = BM - \kappa_C Q^C - \kappa_O Q^O, \quad Q^C = (1 - p_1) e^{-mF(a)} \{\eta_1(a) + \eta_2(a)\} m. \quad (18)$$

That version can ask whether to notify fresh core drivers at all. The current group-meeting comparison instead isolates two questions: first the supply value of replenishment, then the economic scope of additional search.

- Incumbent exit is active through $\omega = .8$ in the main numerics and enters both terminal supply and the focal rejector's waiting value.
- Only the $s = 1$ slice of the expanded-search technology reproduces A; I versus A is explicitly a supply benchmark comparison.
- The peak theorem is deliberately moved out of the main presentation. The next theory step is to characterize the arrival-value peak and the cost-dependent no-search boundary under stable WPBE branches.

Closest modeling languages. Hu, Hu, and Zhu (2022) for post-assignment spatial service costs; Afèche, Liu, and Maglaras (2023) for scarce/intermediate/ample supply regimes; Wang, Zhang, and Zhang (2024) for endogenous search-scope rules.